THOMAS GREEN

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SUMMARY

Robotics engineer with a First-Class MEng in Mechatronic Engineering and an MSc in Robotics, with a dissertation on tactile sensing in vision-language-action (VLA) models. Co-developed and field-tested a patent-pending soft vine robot for fibre-optic installation, moved into production and testing across Liberty Global's operating companies. Currently an AI/Data Engineer at the FIA Formula E World Championship, developing lap-simulation tooling and ML prediction models.

EXPERIENCE

Formula E Sep 2024 – Present

AI/Data Engineer (part-time Oct 2025 – Sep 2026, alongside MSc) *London, UK*

- Built lap-simulation tooling that the track-design team uses to prototype prospective championship circuits, balancing driver safety and energy efficiency against entertainment value. Circuit reviews that previously took weeks of manual work now turn around in a day, even for entirely new tracks with no survey or on-site data.
- Brought race-outcome prediction in-house, building an XGBoost model on engineered features (battery advantage, historical priors, race progression and speed trend) that powers live insights broadcast globally on the world feed. The model calls the winner with 63% accuracy at 60% race distance, up from under 50% for the previous external version.
- Delivered the championship's headline fan-experience product with principal partner Google Cloud: an AI tool that analyses fans' driving on the sim-racing rigs in the Formula E Gaming Arena, used by 10,000+ fans and showcased by Google at conferences worldwide.
- Contributed to the company-wide AWS-to-GCP migration; built real-time data pipelines, rebuilt key visualisation dashboards and wrote internal AI/LLM tools.

Liberty Global Sep 2022 – Aug 2024

Graduate Engineer, Technology Rotational Programme (Innovation, Strategy, AI Automation) *Zurich, Switzerland*

- Co-led ideation and prototyping of a patent-pending soft vine robot that grows through congested underground ducting to install fibre-optic cable; built 3 prototypes in 9 months and demonstrated a single-run traversal of 200+ metres of live duct in field trials. The robot won Group CTO backing for productionisation and moved into production and testing across operating companies.
- Led automation initiatives across Liberty Global subsidiaries, replacing manual workflows with custom Python tooling and freeing capacity equivalent to around 8 full-time roles.

EDUCATION

MSc Robotics Sep 2025 – Sep 2026

University of Bristol, UK *GPA 3.92/4.0*

MEng (Master of Engineering), Mechatronic Engineering Sep 2018 – Aug 2022

University of Southampton, UK *First-Class Honours (GPA 3.9/4.0)*

SELECTED PROJECTS

- **Tactile sensing in a vision-language-action model** (MSc dissertation): built a GelSight-style optical tactile sensor for an ABB YuMi and compared four routes for feeding touch into SmolVLA. With an auxiliary tactile loss, all 24 live paired trials on the robot moved in the correct direction.
- **Humanoid legs from an SO-101 kit** (personal, ongoing): printable legs designed in SolidWorks around 8 STS3215 servos from an SO-101 arm kit, with a MuJoCo model generated directly from the CAD.
- **Smart notebook** (personal, ongoing): designed and hand-soldered a 16-channel piezo amplifier PCB on a Teensy 4.1, streaming 16 channels at 100 kHz to locate a pen by time difference of arrival.
- **Autonomous pedal-actuation robot** (MEng): designed a retrofit actuator that operates a car's pedals, from mechanical design to closed-loop control (114 ms rise, 0.2% overshoot).

TECHNICAL SKILLS

Programming	Python, SQL, MATLAB, C/C++, R, JavaScript
Robotics	Soft Robotics, Robotic Manipulation, Closed-Loop Control, Kinematics, Sensor Integration, Embedded Systems, Machine Vision (OpenCV), MuJoCo, PCB assembly
Machine Learning	PyTorch, scikit-learn, XGBoost, VLA Models (π0.5, SmolVLA, OpenVLA), OpenAI & Claude APIs
CAD & Hardware	SolidWorks, FEA, Rapid Prototyping (SLA/FDM, CNC Machining)
Cloud, Data & Dev	GCP (BigQuery, Vertex AI), AWS, PostgreSQL, Docker, Git, CI/CD, Linux, REST, FastAPI

ADDITIONAL

Availability Available full-time from March 2027 on an IEC open work permit.